PROJECT REPORT

Metro Ticket Generation System Using ServiceNow

Project Name	Metro Ticket Generation System Using ServiceNow
Platform	ServiceNow
Date	2025
Roll No	23A51A4255

1. INTRODUCTION

1.1 Project Overview

The Metro Ticket Generation System is a ServiceNow-based digital solution developed to simplify the metro ticket booking process. The system enables commuters to book metro tickets through a user-friendly Service Catalog interface instead of relying on manual ticket counters. It automates ticket request submission, fare calculation, approval (if required), ticket generation, and record management using ServiceNow workflows.

The project demonstrates how ServiceNow can be used beyond traditional IT Service Management by implementing a real-world public transportation use case. The application improves efficiency, minimizes manual effort, maintains centralized records, and provides a scalable platform for future enhancements.

1.2 Purpose

The purpose of this project is to develop a digital, automated metro ticketing solution on the ServiceNow platform that eliminates manual ticket counter dependency, reduces passenger wait time, and streamlines the entire ticket booking process through a centralized and automated workflow.

2. IDEATION PHASE

2.1 Problem Statement

Traditional metro ticket booking methods often involve long queues, manual processing, and limited digital automation. Managing ticket requests manually increases processing time and makes tracking passenger information difficult. There is a need for a centralized and automated solution that enables passengers to book metro tickets quickly while allowing administrators to manage bookings efficiently.

2.2 Empathy Map Canvas

The Empathy Map for the Metro Ticket Generation System was built around the daily metro commuter persona. The analysis revealed the following user insights:

- THINK & FEEL: Commuters feel frustrated with long queues and time lost at ticket counters. They desire a faster, digital booking experience and worry about missing trains due to ticketing delays.
- HEAR: Commuters hear from peers about digital payment and ticketing options available in other cities. They hear announcements about modernization of public services.
- SEE: Commuters see crowded ticket counters, manual token-based systems, and limited digital infrastructure at metro stations.
- SAY & DO: Commuters express frustration verbally and resort to carrying exact change, arriving early to avoid queues, or avoiding peak hours.
- PAIN: Long queues, manual errors in ticketing, no digital record of travel, time-consuming booking process.
- GAIN: Quick digital booking, automatic fare calculation, centralized booking records, convenience, and time savings.

2.3 Brainstorming

The team conducted a structured brainstorming session focused on the problem of manual metro ticketing. The following ideas were generated and prioritized:

- Idea 1 (Selected): ServiceNow-based Service Catalog for digital ticket booking — High Feasibility, High Impact
- Idea 2: Mobile app for ticket booking — High Impact but high complexity and longer development time
- Idea 3: QR Code kiosk at metro stations — Medium feasibility, requires hardware integration
- Idea 4: WhatsApp chatbot for ticket booking — Medium feasibility, limited to messaging platform

After prioritizing by impact and feasibility, the ServiceNow Service Catalog solution was selected as the primary approach due to its rapid development capability, built-in workflow automation, and scalability.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The Customer Journey Map for the Metro Ticket Generation System traces a commuter's end-to-end experience:

- Awareness: Commuter learns about the digital booking portal through announcements or colleagues.
- Enter: Commuter logs into ServiceNow portal and navigates to the Metro Ticket Booking catalog item.
- Engage: Commuter fills in source station, destination station, travel date, passenger count, and passenger details. Client Scripts validate inputs in real-time.
- Fare Calculation: System automatically calculates fare based on route and passenger count.
- Submission: Commuter submits the request. Flow Designer triggers and creates a booking record.
- Exit: Commuter receives booking confirmation with ticket details stored in the Metro database.
- Extend: Administrator tracks bookings via the Admin Dashboard; commuter can view past bookings.

3.2 Solution Requirement

Functional Requirements:

- FR-1: Metro Ticket Booking — Users can submit a ticket request via Service Catalog with source, destination, date, and passenger details.
- FR-2: Fare Calculation — System auto-calculates fare based on route and number of passengers using Client Scripts.
- FR-3: Input Validation — Client Scripts validate all mandatory fields before form submission.
- FR-4: Booking Record Creation — Flow Designer creates a booking record in the custom Metro database table upon form submission.
- FR-5: Admin Dashboard — Administrators can view, search, and manage all booking records.

Non-Functional Requirements:

- NFR-1: Usability — Simple, intuitive Service Catalog form accessible to all metro commuters.
- NFR-2: Security — Role-based access control; only authorized users can view and manage booking records.
- NFR-3: Reliability — System maintains data integrity through Business Rules and form validation.
- NFR-4: Performance — Service Catalog form loads and submits within acceptable time limits.
- NFR-5: Availability — System is accessible 24/7 via ServiceNow platform.

- NFR-6: Scalability — Architecture supports future enhancements like QR code, payments, and mobile app.

3.3 Data Flow Diagram

Level 0 DFD (Context Diagram): The Metro Ticket Generation System receives travel inputs from the Commuter (source station, destination station, travel date, passenger details) and returns a Booking Confirmation. The Administrator interacts with the system to view and manage booking records in the Metro Database.

Level 1 DFD: Processes include: (1) User Authentication — validates commuter login via ServiceNow; (2) Ticket Request Form — captures and validates passenger inputs; (3) Fare Calculation Engine — computes fare based on route; (4) Flow Designer — triggers booking record creation; (5) Metro Database — stores all booking records; (6) Admin Dashboard — displays booking data to administrators.

3.4 Technology Stack

S.No	Component	Description	Technology
1	User Interface	Service Catalog form for ticket booking	ServiceNow Service Catalog, Form Designer
2	Application Logic	Input validation and fare calculation	Client Scripts, UI Policies
3	Workflow Automation	Automates booking record creation	Flow Designer
4	Backend Logic	Data consistency and backend processing	Business Rules
5	Database	Stores all booking records	ServiceNow Custom Table (Metro DB)
6	Platform / Cloud	Application deployment and hosting	ServiceNow Cloud Platform

4. PROJECT DESIGN

4.1 Problem Solution Fit

The Metro Ticket Generation System achieves a strong Problem-Solution Fit by directly addressing the core pain points of metro commuters:

- Customer Segment: Daily metro commuters who rely on manual ticket counters.
- Problem / Job-to-be-done: Purchase a metro ticket quickly without standing in long queues.
- Available Solutions (Before): Manual token counters, limited kiosks — time-consuming and error-prone.
- Constraints: Lack of digital infrastructure, dependency on cash, no centralized records.

- Our Solution: ServiceNow Service Catalog — enables digital ticket booking with automated fare calculation and centralized record storage.
- Behaviour Match: Commuters already use digital portals for services; ServiceNow mirrors familiar web-form interactions.
- Emotion Before → After: Frustrated by queues → Confident and time-efficient with digital booking.

4.2 Proposed Solution

S.No	Parameter	Description
1	**Problem Statement**	Manual metro ticketing causes long queues, processing delays, and lack of centralized booking records.
2	Idea / Solution Description	A ServiceNow Service Catalog application that allows commuters to digitally book metro tickets with automated fare calculation, workflow-based record creation, and admin management.
3	Novelty / Uniqueness	Uses ServiceNow beyond ITSM for a real-world public transportation use case, demonstrating its versatility for citizen-facing digital services.
4	Social Impact / Customer Satisfaction	Reduces commuter queue time, minimizes manual errors, provides 24/7 digital booking access, and improves overall metro user experience.
5	Business Model	Platform-based digital ticketing service; potential for integration with payment gateways for fare collection and revenue tracking.
6	Scalability	ServiceNow's cloud architecture supports scaling to multiple metro lines, cities, QR code ticketing, mobile app integration, and AI-powered travel assistance.

4.3 Solution Architecture

The Metro Ticket Generation System is built entirely on the ServiceNow platform. The architecture consists of the following layers:

- Presentation Layer: ServiceNow Service Portal / Service Catalog — Metro Ticket Booking form accessible to commuters via web browser.
- Application Logic Layer: Client Scripts (fare calculation, input validation), UI Policies (field visibility control), Business Rules (backend data processing).
- Workflow Layer: Flow Designer — captures catalog variables on form submission and triggers booking record creation automatically.
- Data Layer: Custom ServiceNow Table (Metro Booking Table) — stores all ticket booking records including passenger info, route, fare, and booking status.

- Administration Layer: ServiceNow Admin Dashboard — allows administrators to view, filter, and manage all booking records.
- Future Integration Points: Payment Gateway API, QR Code Generator, SMS/Email Notification, Mobile App.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

S.No	Task / Module	Duration	Status
1	Requirement Analysis & Planning	Week 1	Completed
2	ServiceNow Custom App & Table Setup	Week 2	Completed
3	Service Catalog Item & Form Design	Week 2-3	Completed
4	Client Scripts & UI Policies	Week 3	Completed
5	Flow Designer Automation	Week 4	Completed
6	Business Rules Implementation	Week 4	Completed
7	Testing & Validation	Week 5	Completed
8	Documentation & Presentation	Week 6	Completed

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

TC No	Test Case Description	Expected Result	Status
TC-01	Submit Metro Ticket Booking form with valid inputs	Booking record created in Metro table	Pass
TC-02	Submit form with missing mandatory fields	Client Script shows validation error	Pass
TC-03	Verify automatic fare calculation on route selection	Correct fare displayed before submission	Pass
TC-04	Verify Flow Designer triggers on form submission	Flow executes and booking record is created	Pass
TC-05	Admin views booking records in dashboard	All records displayed with correct details	Pass

TC-06	Business Rule triggers on record creation	Backend processing executes correctly	Pass

7. RESULTS

7.1 Output Screenshots

The following outputs were observed upon successful implementation of the Metro Ticket Generation System:

- Metro Ticket Booking Service Catalog item is accessible via the ServiceNow Service Portal.
- The booking form captures source station, destination station, travel date, number of passengers, and passenger details.
- Client Scripts validate all inputs in real-time and auto-calculate the fare based on the selected route.
- Upon submission, Flow Designer triggers and creates a new record in the custom Metro Booking table.
- The booking record stores all ticket details and is accessible to administrators via the Admin Dashboard.

(Add screenshots of the Service Catalog form, form submission, Flow Designer execution, and Metro database records here.)

8. ADVANTAGES & DISADVANTAGES

Advantages

- Eliminates manual ticket counter dependency, reducing commuter wait time.
- Automated fare calculation ensures accuracy and consistency.
- Centralized Metro database maintains organized booking records for tracking and reporting.
- ServiceNow's built-in security and role-based access control ensures data protection.
- Scalable architecture supports future enhancements like QR code ticketing and payment integration.
- Demonstrates versatile use of ServiceNow beyond traditional ITSM applications.

Disadvantages

- Requires internet access and a ServiceNow account for commuters, limiting accessibility to non-digital users.
- Initial setup and configuration of ServiceNow modules requires technical expertise.
- Currently dependent on manual fare table maintenance for accurate fare calculation.
- No real-time metro schedule integration in the current version.

9. CONCLUSION

The Metro Ticket Generation System successfully demonstrates how the ServiceNow platform can be leveraged beyond traditional IT Service Management to deliver a real-world digital transformation solution. By automating the metro ticket booking process through the Service Catalog, Client Scripts, Flow Designer, and Business Rules, the project significantly reduces manual effort, improves booking accuracy, and maintains centralized travel records.

The system provides a user-friendly, scalable, and secure digital ticketing experience for metro commuters while equipping administrators with a streamlined dashboard for booking management. The project serves as a practical model for implementing digital public services using enterprise platforms like ServiceNow.

10. FUTURE SCOPE

- QR code-based digital ticket generation and validation at metro entry gates.
- Integration with online payment gateways for cashless fare collection.
- AI-powered chatbot for real-time travel assistance and route recommendations.
- Real-time metro schedule and train status integration.
- SMS and email notifications for booking confirmation and travel reminders.
- Mobile application integration for on-the-go ticket booking.
- Route optimization and travel time suggestions.
- Passenger travel history and analytics dashboard.

11. APPENDIX

Source Code

ServiceNow configuration includes: Custom Application Scope, Custom Metro Booking Table, Service Catalog Item with Catalog Variables, Client Scripts for validation and fare calculation, UI Policies for field visibility, Flow Designer workflow for booking record creation, and Business Rules for backend processing.

GitHub & Project Demo Link

GitHub Repository: https://github.com/Vanadharmateja/metro-ticket-generation.git

Project Demo Link: https://drive.google.com/file/d/1HwC7jn_wQIx3bDta-_7euaWISC8d4y-c/view?usp=drivesdk